

Understanding Classification Metrics

Confusion Matrix, Precision, Recall, F1-score, Sensitivity and Specificity

THE EXAMPLE

A model predicts whether a patient has a disease. We compare the model's prediction with the patient's actual result.

The key idea

TRUE The prediction is correct	FALSE The prediction is wrong
POSITIVE Disease	NEGATIVE No disease

A metric simply focuses on a different type of correct or incorrect prediction.

1. Confusion Matrix

A confusion matrix is a table that shows where the model was correct and where it made mistakes.

	PREDICTED: DISEASE	PREDICTED: NO DISEASE
ACTUAL: DISEASE	TRUE POSITIVE (TP) Has disease; model says disease Correct detection	FALSE NEGATIVE (FN) Has disease; model says no disease Missed case
ACTUAL: NO DISEASE	FALSE POSITIVE (FP) No disease; model says disease False alarm	TRUE NEGATIVE (TN) No disease; model says no disease Correct rejection

Example numbers

TP = 40	FP = 10	FN = 5	TN = 50
---------	---------	--------	---------

Total patients = 40 + 10 + 5 + 50 = **105**

2. Precision and Recall

PRECISION

Simple question: When the model predicts disease, how often is it correct?

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 40 / 50 = 80\%$$

Use precision when **false alarms are costly**. Example: marking a genuine email as spam.

RECALL (SENSITIVITY)

Simple question: Out of all patients who truly have the disease, how many did the model find?

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) = 40 / 45 = 88.9\%$$

Use recall when **missing a positive case is dangerous**. Example: failing to detect a serious disease.

Metric	Looks at	Main concern	Memory tip
Precision	Predicted positives	Avoid false alarms	Can I trust a positive prediction?
Recall	Actual positives	Avoid missed cases	Did I find the positives?

Sensitivity is exactly the same as recall.

3. Specificity and F1-score

SPECIFICITY

Simple question: Out of all healthy patients, how many did the model correctly identify as healthy?

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP}) = 50 / 60 = 83.3\%$$

High specificity means the model creates **fewer false alarms** among negative cases.

F1-SCORE

Simple question: How well does the model balance precision and recall?

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) = 84.2\%$$

F1-score is useful when the classes are unequal or when **both false alarms and missed cases matter**.

Quick comparison

Metric	One simple question	Example
Precision	Of predicted positives, how many were correct?	80.0%
Recall / Sensitivity	Of actual positives, how many were found?	88.9%
Specificity	Of actual negatives, how many were identified?	83.3%
F1-score	Is precision and recall balanced?	84.2%

Easy memory trick

Precision Trust the positive prediction	Recall Find the real positives
Specificity Recognize the real negatives	F1-score Balance precision and recall

4. ROC Curve and AUC

The ROC curve shows how a classification model performs at **different decision thresholds**.



HOW TO READ IT

Closer to the top-left
means better performance.

The dotted diagonal line
represents random guessing.

Each point represents a
different threshold.

AUC = Area Under the ROC Curve

It summarizes the entire ROC curve into one score. A higher value generally means the model separates positive and negative cases better.

AUC	Simple interpretation
1.0	Perfect separation
0.9 - 0.99	Excellent
0.8 - 0.89	Good
0.7 - 0.79	Fair
0.5	Similar to random guessing

Easy memory tip: ROC shows performance across thresholds; AUC gives one overall score.